Pouya Hosseinzadeh

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Google Scholar Website LinkedIn GitHub

Education

Ph.D. in Computer Science, *Utah State University, USA* Jan. 2022 – Dec. 2026 (expected)
Advisor: Prof. Soukaina Filali Boubrahimi
Main research: Machine learning for time-series, multimodal, and structured data; explainable AI; representation learning; data-centric learning; graph-based learning; applications in heliophysics, hydrology, and environmental science
GPA: 3.8/4

M.Sc. in Computer Engineering, *University of Siena, Italy* Sep. 2017 – Jul. 2021
Advisor: Prof. Marco Maggini
Thesis: Comparison of the Statistics of Human and Automatically Generated Texts
GPA: 3.7/4

B.Sc. in Computer Engineering, *University of Tabriz, Iran* Sep. 2011 – Feb. 2016
Specialization: Computer Architecture

Honors and Grants

National Science Foundation (NSF) Funded Research Projects, *Utah State University* 2022 – Present
Contributed to multiple NSF-funded projects in space weather forecasting and heliophysics, including:

- Combining Physics and Machine Learning-based Models for Full-Energy-Range Solar Energetic Particle Event Prediction (Award #2204363)
- CAREER: End-to-End Active Region-based Heliospheric Forecasting System Using Multi-spacecraft Data and Machine Learning (Award #2240022)
- SHINE: Multimodal Machine Learning Approaches for Solar Energetic Particle Event Prediction and Posthoc Analysis (Award #2501486)

Graduate Research Assistantship (GRA), *Utah State University* 2022 – Present
Fully funded Ph.D. position supporting research in space weather forecasting and machine learning

Research Experience and Projects

Doctoral Researcher Jan. 2022 – Present
Utah State University, USA

- Conduct research in **machine learning for temporal, multimodal, and structured data, including explainable AI, representation learning, data augmentation, data condensation, graph-based learning, and scientific data analysis**.
- Conducted research on **space weather and heliophysics**, focusing on **solar energetic particle (SEP)** and **solar flare prediction** using observational data and machine learning.
- Published first-author studies in *The Astrophysical Journal Supplement Series (ApJS)* and *Space Weather* on multivariate time series modeling, multimodal data fusion, and ensemble learning for **high-impact SEP event prediction**.
- Developed forecasting pipelines using observational heliophysics datasets, including **GOES** (X-ray and proton flux), **NOAA** event catalogs, and **NASA OMNI** solar wind and magnetic field data.
- Contributed to solar flare forecasting and feature analysis, including time-series feature selection methods for prediction. [DOI]
- Developed **explainable AI** and time-series learning methods, including **counterfactual explanation** models, **multimodal classification** frameworks, and **data augmentation techniques**.

- Conducted related research in hydrology and **environmental forecasting**, including streamflow prediction using **spatio-temporal models** and satellite data. [DOI]

Teaching and Mentorship Experience

Graduate Teaching Assistant, *Utah State University* — Jan. 2022 - May 2022
Introduction to Computer Science

Guest Lecturer, *Utah State University* — 2023
Applied Deep Learning (Mar. 2023), Time Series Data Mining (Nov. 2023)

Graduate Research Mentor, Utah State University — 2023 – Present

- Supervised 7 M.S. students on machine learning research projects in SEP prediction, solar flare forecasting, time-series analysis, and hydrology modeling, with several resulting in peer-reviewed publications.
- Helped manage the Data Mining Lab and mentored Ph.D. students in the early stages of their research projects.

Presentations and Talks

SHINE Conference (Solar Heliospheric and INterplanetary Environment), Juneau, AK — Aug. 2025
Hosseinzadeh, P. "Solar Energetic Particle Event Prediction using Multivariate Time Series and Machine Learning" (oral)

NASA SEP Machine Learning Monthly Meeting — Aug. 2024
Hosseinzadeh, P. "Machine Learning Approaches for Solar Energetic Particle Prediction" (invited talk)

International Conference on Machine Learning and Applications (ICMLA), Jacksonville, FL — Dec. 2023
Hosseinzadeh, P. et al. "Metforc: Classification with Meta-learning and Multimodal Stratified Time Series Forest" (oral)

AGU Fall Meeting, Chicago, IL — Dec. 2022
Hosseinzadeh, P., Nassar, A., Filali Boubrahimi, S. "AI-Based Streamflow Prediction in the Upper Colorado River Basin Using Multimodal Spatiotemporal Data" (poster)

Professional Service

Conference Reviewer: NeurIPS, UAI, IJCAI, AISTATS, ICMLA
Journal Reviewer: The Astrophysical Journal Supplement Series (ApJS), Space Weather, Advances in Space Research, Scientific Reports (Nature Portfolio), Pattern Recognition (Elsevier)

Technical Skills

Programming: Python, MATLAB, Java
Machine Learning: Time-series modeling, deep learning, multimodal learning, representation learning, graph neural networks, explainable AI, data augmentation, data condensation
Frameworks: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas
Scientific Domains: Space weather, heliophysics, hydrology, environmental science
Data Analysis: Multivariate time-series analysis, statistical modeling, data visualization

Selected Publications

Hosseinzadeh, P., Li, P., Filali Boubrahimi, S., & Hamdi, S. M. (2026, August). TopGrad-CF: Gradient-Guided Counterfactual Explanations for Time Series Classification. In *International Conference on Pattern Recognition* (pp. 512-526). Cham: Springer Nature Switzerland. DOI

Li, P., **Hosseinzadeh, P.**, Filali Boubrahimi, S., & Hamdi, S. M. (2026). RoboCF: Diffusion-Guided Robust Counterfactual Explanations for Time Series Classification. In *IEEE International Conference on Data Mining (ICDM)*. Accepted.

Hosseinzadeh, P., Filali Boubrahimi, S., & Hamdi, S. M. (2025). An End-to-end Ensemble Machine Learning

Approach for Predicting High-impact Solar Energetic Particle Events Using Multimodal Data. *The Astrophysical Journal Supplement Series (ApJS)*, 277(2), 34. DOI

Kasapis, S., **Hosseinzadeh, P.**, Whitman, K., Egeland, R., Georgoulis, M., Vourlidas, A., Papaioannou, A., Lavasa, E., Anastasiadis, A., et al. (2026). Review of Machine Learning Models for Solar Energetic Particle Prediction. *Space Science Reviews (SSR)*. Under Review.

Hosseinzadeh, P., Li, P., Bahri, O., Filali Boubrahimi, S., & Hamdi, S. M. (2025). CACTUS: Cross-Aligned Counterfactual Explanation for Time Series Classification. In *IEEE International Conference on Data Science and Advanced Analytics (DSAA)*. DOI

Hosseinzadeh, P., Filali Boubrahimi, S., & Hamdi, S. M. (2024). Improving Solar Energetic Particle Event Prediction through Multivariate Time Series Data Augmentation. *The Astrophysical Journal Supplement Series (ApJS)*. DOI

Hosseinzadeh, P., Filali Boubrahimi, S., & Hamdi, S. M. (2024). Toward Enhanced Prediction of High-Impact Solar Energetic Particle Events Using Multimodal Time Series Data Fusion Models. *Space Weather*. DOI

Velanki, Y., **Hosseinzadeh, P.**, Filali Boubrahimi, S. F., & Hamdi, S. M. (2024). Time-series feature selection for solar flare forecasting. *Universe*, 10(9), 373. DOI

Whitman, K., Egeland, R., Richardson, I. G., Allison, C., Quinn, P., Barzilla, J., **Hosseinzadeh, P.**, et al. (2023). Review of Solar Energetic Particle Prediction Models. *Advances in Space Research*, 72(12), 5161–5242. DOI

Hosseinzadeh, P., Li, P., Bahri, O., Boubrahimi, S. F., & Hamdi, S. M. (2026). TexTSC: Class-Texture Preserving Data Condensation for Time Series Classification. *International Conference on Artificial Intelligence and Statistics (AISTATS)*. DOI

Filali Boubrahimi, S., Neema, A., Nassar, A., **Hosseinzadeh, P.**, & Hamdi, S. M. (2024). Spatiotemporal Data Augmentation of MODIS-Landsat Water Bodies Using Adversarial Networks. *Water Resources Research*, 60(3), e2023WR036342. DOI

Hosseinzadeh, P., Li, P., Bahri, O., Filali Boubrahimi, S., & Hamdi, S. M. (2024). ACTS: Adaptive Counterfactual Explanations for Time Series Data Using Barycenters. In *IEEE International Conference on Big Data (BigData)*. DOI

Akkala, A., Boubrahimi, S. F., Hamdi, S. M., **Hosseinzadeh, P.**, & Nassar, A. (2025). Spatio-Temporal Graph Neural Networks for Streamflow Prediction in the Upper Colorado Basin. *Hydrology*, 12(3), 60. DOI

Hosseinzadeh, P., Nassar, A., Boubrahimi, S. F., & Hamdi, S. M. (2023). ML-Based Streamflow Prediction in the Upper Colorado River Basin Using Climate Variables Time Series Data. *Hydrology*, 10(2), 29. DOI